From Feature Spec to Insight in Seconds

An Autonomous Multi-Agent Analytics System Built on ClickHouse

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The Product Velocity Bottleneck

Atlys ships fast across 120+ visa destinations — but every feature triggers a manual cascade: PRDs, hand-written DDL, custom SQL. Context dies in handoffs. Base layers carry silent contradictions.

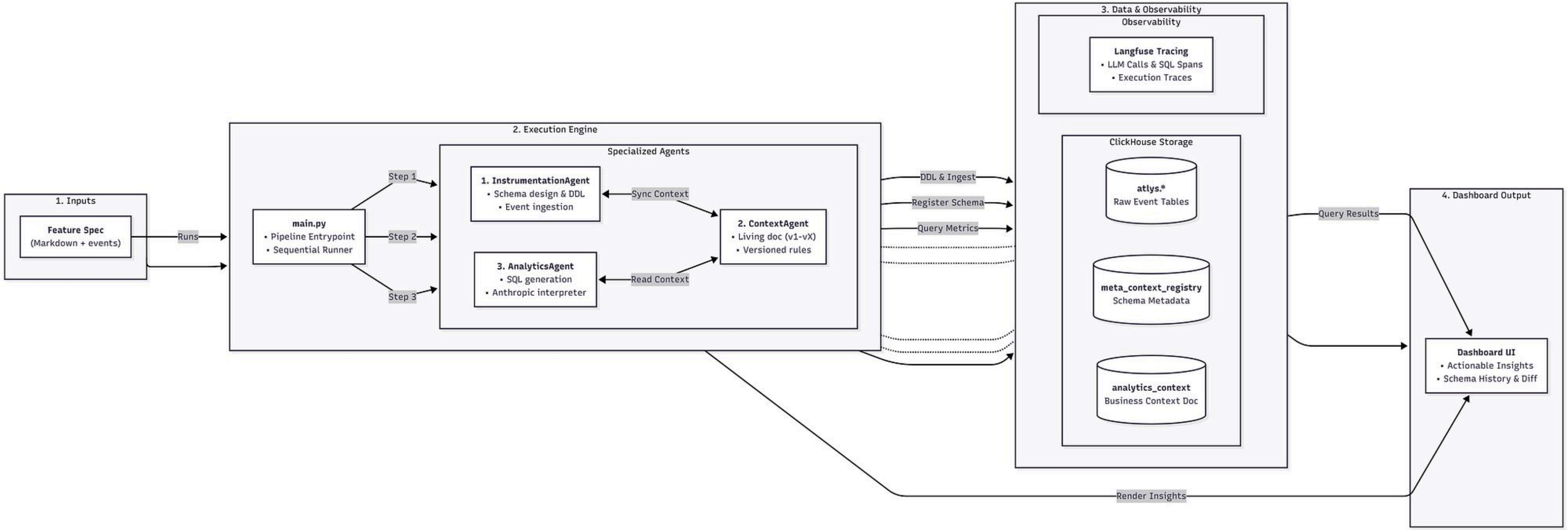
Old Manual Flow

PRD → Manual DDL (weeks later) → Stale Dashboard

Our Agentic Pipeline

Feature Spec JSON → Instrumentation → Living Context → Grounded Insights in seconds

High-Level Architecture



Made with GAMMA

Our Core Technology Stack

Python Agents

The brains of our operation. All intelligent agents are developed in Python, leveraging its robust ecosystem for AI, data processing, and automation logic.

Streamlit UI

An intuitive and interactive user interface built with Streamlit, enabling seamless interaction with the agent system and real-time data visualization.

ClickHouse DB & Querying

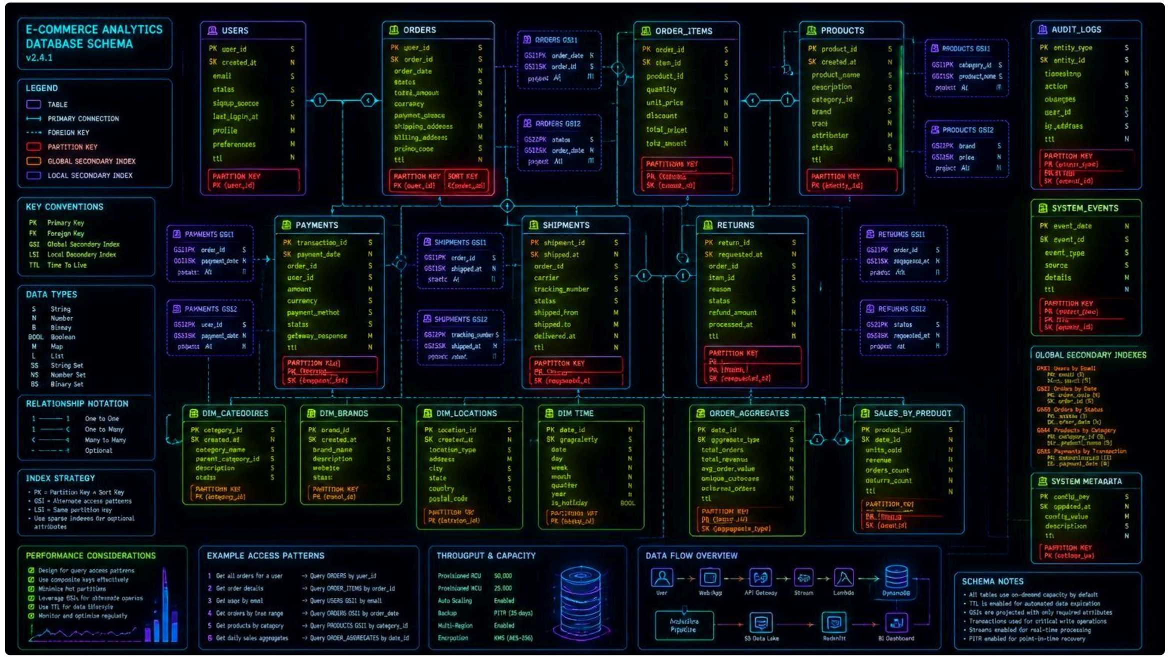
A powerful, high-performance columnar database for real-time analytics. ClickHouse serves as our central data store and analytical engine, ensuring rapid query responses.

Langfuse Observability

Integrated for end-to-end observability of our LLM-powered applications. Langfuse provides invaluable insights into agent traces, evaluations, and performance metrics.

AGENT 1

Instrumentation Agent — Optimized DDL Generation



Takes raw feature specs and JSON events, outputs **production-grade ClickHouse schemas** — not generic syntax along with context which is feeded in.

AGENT 2

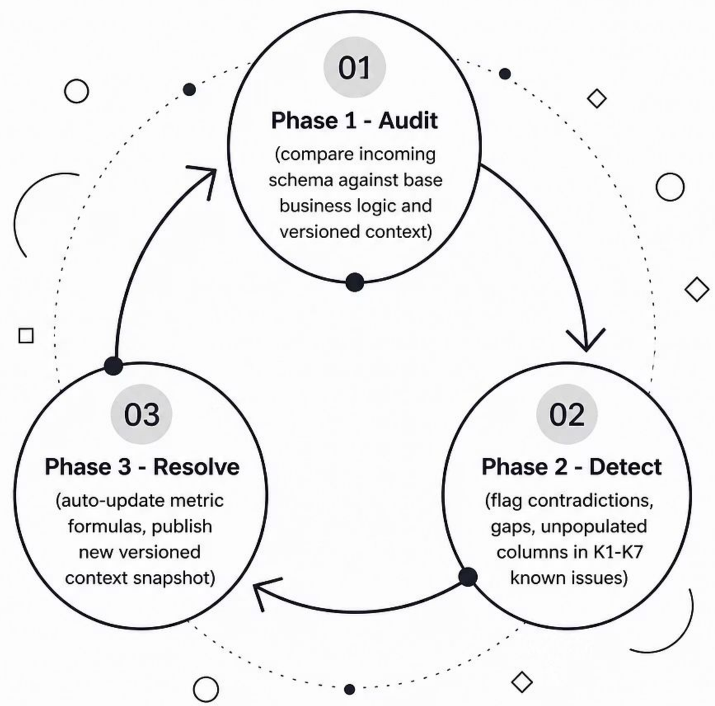
Context Agent — The Living Context Engine

Storage: Native Versioned Row in ClickHouse

Context versioning resides **directly alongside the data warehouse** — no external sync dependencies. Each update is a incremented version_id snapshot.

Self-Healing & Contradiction Resolution

- Audits incoming schemas against base business logic
- Auto-flags context gaps and contradictions across K_1–K_7 known issues
- Auto-updates metric formulas so Analytics Agent always queries the newest snapshot



AGENT 3

Analytics Agent — Grounded NL-to-SQL

Token Budget Strategy

The agent **never** fetches raw rows into the LLM context. All computation is pushed natively to ClickHouse using `windowFunnel`, `countIf`, `avgIf`, and `coalesce`.

Translating "What" into "Why"

Synthesizes SQL output with the analytics agent to explain root causes — e.g., linking a drop from `pay_now_clicked → purchase_completed` on iOS directly to **K1: iOS WebKit OTP autofill regression**.

The Unseen Spec Pipeline — Proof of Execution

Built for Zero-Shot Generalization

Spec-agnostic parsing ingested the surprise **6th Spec instantly** — no prompt hardcoding, no schema assumptions.

Validation Checkpoint

No Trace, No Credit

Every action — from JSON parsing to DDL execution and final insight generation — is **fully logged and auditable** in Langfuse.

01

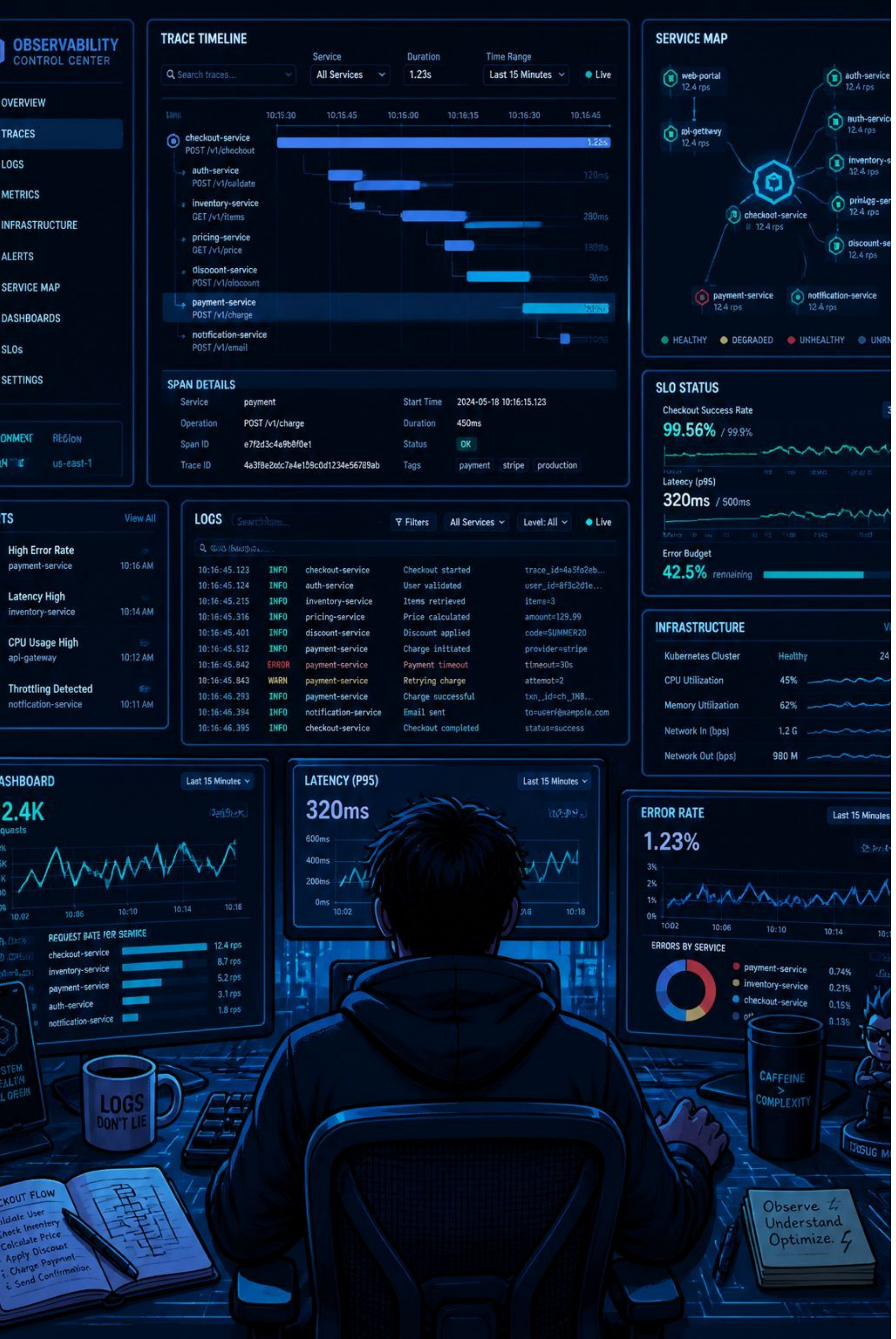
Auto-generated DDL executed against ClickHouse

02

Context diff auto-generated and versioned

03

Product-oriented insight summary produced



Observability & Tracing — Langfuse Integration

Agent-Level Tracing

Full chain-of-thought visibility: *what* the agent did, *why*, and *which versioned context snapshot* it relied upon. Latency and token consumption measured per call.

Executive Insights View

Direct product recommendations accompanied by confidence scores — grounded in verified ClickHouse aggregations, not LLM hallucinations.

Evaluation Summary & Future Roadmap

Hackathon Evaluation Alignment

Schema Quality

Optimized DDL: SharedMergeTree, strict sorting keys, zero unnecessary blobs

Insight Quality

Clear "Why it happened" explanations tied to business context

Context Freshness

Real-time sync between schema evolution and context snapshotting

Future Roadmap

1. **Langfuse Online Evals**
2. **Human-in-the-Loop Webhooks**
3. **Multi-Spec Parallel Ingestion**